An open ecosystem for growing FAIR and tidy data

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Problem

- Scientific environments often lag behind industry in adopting modern technological and engineering best practices
- Research software tooling is often fragmented or lacks maturity
- Funding for open-source software in research remains limited
- Many researchers and funders are not aware of the power and value of robust sotware engineering expertise and tools
- Data storage, processing, and management are underprioritised despite their direct influence on data usability
- **Consequence: Limited capacity to produce and maintain FAIR (findable, accessible, interoperable, reusable), high-quality datasets**

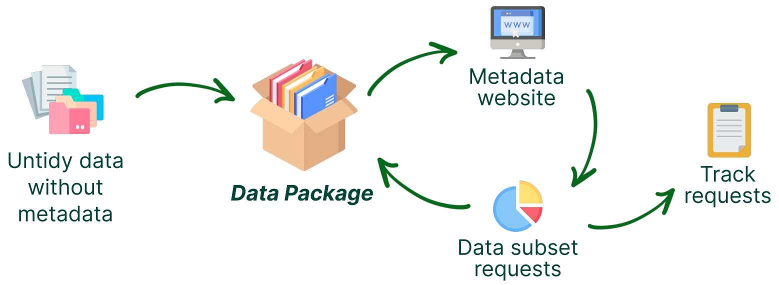
Solution

At a systems level:

- Investment in research software
- Building open source tools for scale and easier use
- Improvement of tooling for tidy, FAIR data workflows

The Seedcase Project:

- Building an open-source ecosystem for engineering and managing high-quality and FAIR research data
- Strong focus on being easy to use by people with various technical backgrounds



Products

Built

Sprout: Grow organised and FAIR data

Flower: Turn your metadata into human-readable documentation

check-datapackage: Ensure the compliance of your Data Package metadata

Template Data Package: An opinionated template for building Data Packages using Seedcase

Planned

Propagate: Submit requests for Data Package data

Garden: Track and manage Data Package requests

Collaborators and users

ON-LiMiT: Large intervention study examining remission of type 2 diabetes through diet and exercise in Denmark
onlimit.org

DP-Next: Sustainable Type 2 Diabetes Prevention for the 21st Century
dp-next.github.io

Steno Diabetes Center Aarhus: Helping people with or at risk of diabetes, through treatment and research
steno-aarhus.github.io

Values

- Build and share openly
- Use other open-source tools
- Documentation-driven
- Beginner-friendly

Platform

Tools

t-squared: Template for building and managing templates

Template Website: Set up and manage website projects

Template Python Package: An opinionated toolkit for building Python packages

zen-do: Command line utilities for working with Zenodo

Soil: A shared foundation for building our Python packages

Documentation

Design: Principles and patterns for development

Community: Community building, outreach, and updates

Team: Internal docs for onboarding and admin information

Guidebook: A guidebook for team collaboration on GitHub

Decisions: A record of decisions on tools and processes

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